Notes: Hirshleifer, Peng & Wang (RFS, 2025)

News Diffusion in Social Networks and Stock Market Reactions

Contents

1 Big picture

- **Research question.** How does the social transmission of public news affect investor attention, beliefs, trading, and stock-price efficiency?
- **Empirical setting.** The paper studies corporate earnings announcements and asks whether announcements by firms headquartered in more socially central counties are incorporated into prices faster.
- **Main network variable.** Firm centrality is the centrality of the firm headquarters county in the Facebook Social Connectedness Index network across U.S. counties.
- **Core hypothesis.** Earnings news first attracts local attention and then spreads through social ties. A firm whose local investor base sits in a more central county should have faster news diffusion.
- **Immediate price result.** High-centrality firms have stronger immediate price reactions to earnings surprises during the $[0, 1]$ announcement window.
- **Post-announcement price result.** High-centrality firms have weaker post-earnings-announcement drift, consistent with more timely price discovery.
- **Volatility result.** High-centrality firms experience stronger immediate uncertainty resolution and faster decay of return volatility.
- **Volume puzzle.** Trading volume does not decay faster. Instead, high-centrality firms have higher and more persistent post-announcement volume.
- **Social churning hypothesis.** Social networks improve information diffusion but also generate continuing discussion, disagreement, attention, and excessive trading.
- **Mechanism evidence.** StockTwits messages, Google searches, and household account data confirm that centrality increases discussion, disagreement, attention, trading, trading costs, and losses.
- **Economic takeaway.** Social networks can make prices more efficient while simultaneously making trading less efficient at the investor level.

2 Data and measurement

2.1 Earnings-announcement sample

- The main sample consists of U.S. common stocks traded on NYSE, AMEX, NASDAQ, and NYSE Arca.
- Historical headquarters locations are taken from SEC EDGAR 10-K header files.
- Quarterly earnings and forecasts come from Compustat and I/B/E/S; stock returns and trading data come from CRSP; accounting controls come from CRSP-Compustat.
- The final main sample contains 238,195 firm-quarter observations from 1996 through 2017.

Interpretation. Headquarters location identifies the local investor base likely to notice the firm first. The empirical question is whether the social-network position of that local base changes how quickly earnings news spreads beyond local investors.

2.2 Facebook Social Connectedness Index and centrality

- The Social Connectedness Index (SCI) measures friendship links between U.S. counties using Facebook data.
- The paper forms a weighted adjacency matrix $S = \{s_{ij}\}$, where s_{ij} is the SCI connection between counties i and j.

- A firm's centrality is the centrality of its headquarters county in this county network.
- The three main centrality measures are degree centrality, eigenvector centrality, and information centrality.
- Degree centrality captures direct connectedness; eigenvector centrality gives more weight to connections to other central nodes; information centrality uses the network's information paths.

Interpretation. Centrality is not just geography or firm size. The heat map shows nearby counties can have sharply different network centrality, which helps separate social connectedness from physical proximity.

2.3 Earnings surprises and market outcomes

- **SUE.** Standardized unexpected earnings is actual EPS minus same-quarter EPS one year earlier, divided by the historical standard deviation of that difference, then converted to a decile rank.
- $CAR[0,1]$**.** Announcement-window abnormal return over days 0 and 1.
- $CAR[2,61^*]$**.** Post-announcement abnormal return from day 2 through five days before the next announcement.
- **LNVOL.** Abnormal log trading volume relative to the preannouncement $[-41,-11]$ window.
- $d_{|R|}$**.** Volatility persistence parameter estimated from an ARFIMA model for daily absolute returns over $[0,61^*]$.
- d_{VOL}**.** Volume persistence parameter estimated from abnormal log volume over $[0,61^*]$.

Interpretation. Returns measure price discovery; volatility decay measures uncertainty resolution; volume measures continuing trading and belief churn.

2.4 StockTwits, Google Search, and household data

- The StockTwits sample uses more than 10 million messages posted by 79,176 unique users from 2009 to 2016.
- New messages represent original attention to a stock; replies represent continuing discussion.
- Disagreement is measured with sentiment from StockTwits messages and decomposed into message-level and reply-level disagreement.
- Influencer centrality is based on the number of StockTwits followers of users who post about the stock.
- Google Search volume is used as a broader measure of investor attention.
- Household account data from a large U.S. discount broker provide 3.9 million announcement-household observations from 1992 to 1996.

Interpretation. These auxiliary datasets test the mechanism. Facebook centrality is broad and persistent; StockTwits and Google reveal real-time attention and opinion dynamics; household data reveal actual trading and trading losses.

3 Models and empirical specifications

3.1 Return reaction regression

$$CAR_{it} = \alpha + \beta_1 SUE_{it} + \beta_2 (CEN_i \times SUE_{it}) + \beta_3 CEN_i + \gamma' X_{it} + \varepsilon_{it}. \tag{1}$$

- CAR is either $CAR[0,1]$ or $CAR[2,61^*]$.

- CEN is one of the three centrality measures.
- The coefficient of interest is β_2, the extra price response to an earnings surprise for high-centrality firms.
- Controls include firm characteristics, county characteristics, industry fixed effects, time fixed effects, and interactions of controls with SUE.

Interpretation. A positive β_2 for $CAR[0,1]$ means centrality makes the immediate price response to earnings news stronger. A negative β_2 for $CAR[2,61^*]$ means centrality makes post-announcement drift weaker.

3.2 Volatility persistence regression

$$d_{|R|,it} = \alpha + \beta_1 CEN_i + \beta_2 |SUE|_{it} + \gamma' X_{it} + \varepsilon_{it}. \tag{2}$$

Interpretation. If social networks diffuse news quickly, then uncertainty should be resolved more quickly and volatility persistence should fall.

3.3 Trading-volume regression

$$LNVOL_{it} = \alpha + \beta_1 CEN_i + \beta_2 |SUE|_{it} + \gamma' X_{it} + \varepsilon_{it}. \tag{3}$$

Interpretation. The volume results distinguish simple information diffusion from social churning. Simple diffusion predicts fast decay; social churning predicts persistent trading.

3.4 StockTwits message and disagreement regressions

$$Y_{it} = \alpha + \beta_1 CEN_i + \beta_2 |SUE|_{it} + \gamma' X_{it} + \varepsilon_{it}. \tag{4}$$

Y is abnormal new messages, abnormal replies, disagreement, or disagreement persistence.

Interpretation. Centrality should increase initial attention but may reduce new-message activity later as the pool of uninformed investors is exhausted. Replies and disagreement can remain high because social discussion generates continued belief updating and argument.

3.5 Influencer regression

$$ARM[2,61^*]_{it} = \beta_1 INFL[0,1]_{it} + \beta_2 ANM[0,1]_{it} + \beta_3 |SUE|_{it} + \gamma' X_{it} + \varepsilon_{it}. \tag{5}$$

Interpretation. If influential users spread news more broadly, their initial posts should lead to more later replies, more disagreement, faster volatility decay, and more persistent volume.

3.6 Household trading regression

$$Trade_{ijt} = \alpha + \beta_1 RSCI_{ij} + \beta_2 |SUE|_{it} + \gamma' X_{it} + \eta' Z_{jt} + \varepsilon_{ijt}. \tag{6}$$

Interpretation. This specification asks whether stronger social ties from the household's county to the firm's county translate into more household trading and worse household performance after earnings news.

4 Identification and empirical design

4.1 What identifies the main relation

- The key variation is cross-sectional: firms headquartered in socially central counties are compared with firms headquartered in less central counties.
- Event timing around earnings announcements helps isolate news-response behavior.
- SUE interactions test whether centrality changes the sensitivity of prices to the news content, not only unconditional returns.
- Industry and time fixed effects absorb broad common shocks and sector-level patterns.

4.2 Why the design is credible

- The Facebook network is measured independently of firms' earnings announcements.
- County centrality has substantial within-state and neighboring-county variation.
- The correlation of centrality with most firm characteristics is modest.
- The paper controls for firm size, book-to-market, earnings persistence, earnings volatility, idiosyncratic volatility, reporting lag, institutional ownership, attention proxies, and county demographics.
- Oster-style omitted-variable tests assess how strong unobservable selection would need to be.

Interpretation. The paper cannot randomly assign counties to different network positions, but it works hard to separate network centrality from firm visibility, geography, and local demographics.

4.3 Mechanism triangulation

- StockTwits shows centrality is associated with more discussion and disagreement.
- Google Search shows centrality is associated with broader retail attention.
- Household trading data show social ties translate into actual trades and losses.
- StockTwits centrality is compared with Facebook centrality to distinguish broad social networks from platform-specific online activity.

Interpretation. The auxiliary tests support the social churning mechanism: centrality both accelerates diffusion and creates persistent belief disagreement and trading.

4.4 What the paper cannot fully prove

- The paper does not observe all private conversations about earnings news.
- Facebook centrality is a static snapshot and may not perfectly represent the social network in every year.
- Firms may choose headquarters locations for reasons correlated with investor attention.
- StockTwits users and discount-broker households are not representative of all investors.
- The results document strong associations and mechanism-consistent patterns, not a randomized experiment.

5 Tables and figures

The figures and tables below are directly cropped from the original paper. Adjacent visuals are grouped to save space and improve reading speed.

5.1 Figure 1 and Table 1: Social centrality and baseline variables

Read:

- Figure 1 shows strong geographic variation in eigenvector centrality across firm-headquarters counties.
- Table 1 shows that centrality measures are highly correlated with each other but only modestly correlated with most firm characteristics.
- The correlation between centrality and firm size is small, helping distinguish social centrality from firm visibility.

Economic message. The network variable is plausibly a distinct information-flow channel rather than a simple proxy for large firms or large metro areas.

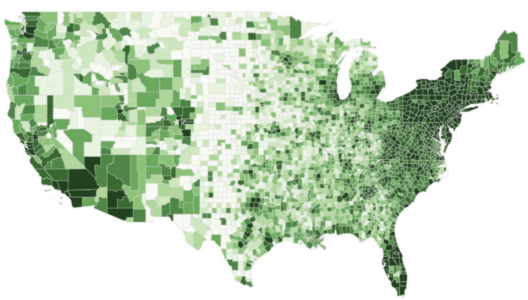

Figure 1
Heat map of eigenvector centrality
This heat map shows eigenvector centrality across U.S. counties that serve as headquarters for publicly listed firms as of June 2016. Darker colors represent higher centrality value deciles. The 10 counties with the highest eigenvector centrality are Los Angeles (CA), Cook (IL), Orange (CA), San Bernardino (CA), San Diego (CA), Riverside (CA), Maricopa (AZ), New York (NY), Clark (NV), and Harris (TX). The 10 counties with the lowest eigenvector centrality are King (TX), McPherson (NE), Wheeler (NB), Slope (ND), Sioux (NE), Blaine (NE),

(a) Figure 1: Heat map of eigenvector centrality

Table 1
Descriptive statistics

A. Descriptive statistics

					Percentile			
Variable	Mean	Median	Stdev	Skewness	10th	25th	75th	90th
DC	18.84	13.14	21.73	2.29	2.11	6.01	20.85	40.15
EC	4.76	0.47	17.91	5.02	0.04	0.17	1.78	5.14
IC	97.90	99.26	4.62	−5.42	95.34	98.42	99.61	99.90
SUE	0.29	0.19	1.36	0.46	−1.41	−0.49	1.02	1.97
CAR[0, 1] (%)	0.02	−0.11	8.91	1.78	−8.81	−3.64	3.49	8.69
CAR[2, 61*] (%)	−0.74	−1.73	26.98	12.23	−23.95	−11.69	7.88	20.24
LNVOL[0, 1]	0.64	0.61	0.99	−0.04	−0.38	0.13	1.14	1.75
LNVOL[2, 61*]	0.04	0.02	0.59	0.35	−0.61	−0.27	0.32	0.70
Size	3.58	0.34	17.60	0.00	0.03	0.09	1.42	5.61
B/M	0.65	0.53	0.47	1.19	0.16	0.30	0.87	1.34
EP	0.17	0.12	0.43	0.34	−0.34	−0.13	0.46	0.76
EVOL	0.86	0.14	4.07	8.65	0.03	0.06	0.35	0.95
IVOL	0.03	0.02	0.02	1.95	0.01	0.01	0.03	0.05
RL	33.65	30.00	16.99	4.59	18.00	23.00	40.00	50.00
IO	0.50	0.51	0.31	0.15	0.07	0.22	0.76	0.91
ADX	30.60	0.00	233.70	17.34	0.00	0.00	0.91	18.05
NA	219	204	136	0.61	46	111	304	420
WSI	0.09	0.08	0.06	1.37	0.03	0.04	0.12	0.17
AvgAge	37.03	36.65	3.37	0.64	33.10	34.57	39.15	41.42
Retire	0.14	0.13	0.04	1.32	0.09	0.11	0.16	0.19
Edu	13.32	13.34	0.68	−0.20	12.50	12.83	13.83	14.17
Income	54.50	51.88	19.07	0.00	32.24	42.24	65.89	80.94
PopDen	4647	1510	13356	4	237	676	2411	5452
Tenancy	7.17	7.00	2.49	0.34	4.00	5.39	9.00	10.00
SPC	13.61	13.41	1.04	0.71	12.43	12.94	14.15	14.96

(Continued)

(b) Table 1, Panel A: Descriptive statistics

5.2 Table 2: Centrality and return reactions to earnings news

Read:

- Panel A shows a positive $CEN \times SUE$ coefficient for $CAR[0,1]$ across all centrality measures.
- In full specifications, the $CEN \times SUE$ coefficients are 0.0152, 0.0149, and 0.0172.
- Panels B and C show negative $CEN \times SUE$ coefficients for post-announcement return windows.

Economic message. Centrality accelerates price discovery: prices react more immediately to earnings news and drift less afterward.

Table 1
(Continued)

B. Correlation structure

	DC	EC	IC
DC	1.000		
EC	0.875	1.000	
IC	0.969	0.902	1.000
SUE	−0.035	−0.046	−0.036
CAR[0, 1] (%)	−0.005	−0.004	−0.005
CAR[2, 61*] (%)	−0.006	−0.005	−0.006
LNVOL[0, 1]	0.005	0.023	0.008
LNVOL[2, 61*]	0.004	0.005	0.005
Size	0.062	0.033	0.057
B/M	−0.036	−0.093	−0.056
EP	−0.019	0.012	−0.013
EVOL	−0.017	−0.021	−0.013
IVOL	0.022	0.073	0.034
RL	0.037	0.039	0.049
IO	0.014	−0.007	0.009
ADX	0.052	0.039	0.064
NA	0.024	0.034	0.029
WSI	−0.169	−0.100	−0.194
AvgAge	−0.245	−0.211	−0.225
Retire	−0.257	−0.317	−0.281
Edu	−0.165	−0.028	−0.109
Income	−0.063	−0.059	−0.050
PopDen	0.309	0.313	0.353
Tenancy	−0.248	−0.210	−0.270
SPC	0.360	0.349	0.426

This table reports the summary statistics and correlation matrix for the main variables used in the paper. Panel A reports the mean, median, standard deviation, skewness, and the 10th, 25th, 75th, and 90th percentiles for each variable. The centrality measures, degree centrality (DC), eigenvector centrality (EC), and information centrality (IC) are scaled so that the maximum value of each is 100. Panel B reports the time-series averages for the cross-sectional correlations of the decile ranks of the centrality measures against other variables. Variable descriptions are in Table A1 in the appendix.

Table 2
Centrality and returns following earnings announcements

A. CAR[0, 1]

	Degree centrality			Eigenvector centrality			Information centrality		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CEN·SUE	0.00737***	0.00673**	0.0152***	0.00766***	0.00635**	0.0149***	0.00801***	0.00685**	0.0172***
	(2.78)	(2.42)	(4.68)	(2.90)	(2.29)	(4.39)	(3.02)	(2.45)	(5.06)
SUE	0.405***	0.423***	1.386***	0.403***	0.425***	1.428***	0.402***	0.422***	1.413***
	(24.89)	(24.52)	(5.26)	(24.76)	(24.71)	(5.42)	(24.90)	(24.63)	(5.39)
CEN	−0.0558***	−0.0430**	−0.0909***	−0.0723***	−0.0440***	−0.0933***	−0.0620***	−0.0412**	−0.0998***
	(−3.68)	(−2.51)	(−4.81)	(−4.76)	(−2.58)	(−4.81)	(−4.07)	(−2.38)	(−5.07)
Ctrls		X	X		X	X		X	X
SUE·Ctrls			X			X			X
Obs.	253,148	226,986	226,986	253,148	226,986	226,986	253,148	226,986	226,986
Adj. R^2 (%)	2.1	2.5	3.2	2.1	2.5	3.2	2.1	2.5	3.2

B. CAR[2, 40]

	Degree centrality			Eigenvector centrality			Information centrality		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CEN·SUE	−0.0180***	−0.0206***	−0.0123**	−0.0218***	−0.0238***	−0.0142**	−0.0166***	−0.0187***	−0.00997
	(−3.70)	(−4.10)	(−2.03)	(−4.48)	(−4.73)	(−2.29)	(−3.44)	(−3.74)	(−1.60)
SUE	0.390***	0.400***	0.201	0.412***	0.419***	0.201	0.382***	0.390***	0.118
	(12.73)	(12.55)	(0.36)	(13.91)	(13.58)	(0.37)	(12.81)	(12.56)	(0.21)
CEN	0.159***	0.163***	0.116***	0.215***	0.207***	0.151***	0.153***	0.154***	0.104**
	(4.75)	(4.61)	(2.88)	(6.15)	(5.70)	(3.59)	(4.53)	(4.32)	(2.49)
Ctrls		X	X		X	X		X	X
SUE·Ctrls			X			X			X
Obs.	252,184	226,106	226,106	252,184	226,106	226,106	252,184	226,106	226,106
Adj. R^2 (%)	0.2	0.4	0.6	0.2	0.4	0.6	0.2	0.4	0.6

(Continued)

Figure 2: Table 2, Panels A–B: Announcement returns and medium-window drift

Table 2
(Continued)

	C. CAR[2, 61]*								
	Degree centrality			Eigenvector centrality			Information centrality		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CEN·SUE	−0.0213***	−0.0227***	−0.00994	−0.0274***	−0.0292***	−0.0141*	−0.0203***	−0.0213***	−0.00726
	(−3.35)	(−3.40)	(−1.27)	(−4.12)	(−4.22)	(−1.77)	(−3.20)	(−3.18)	(−0.90)
SUE	0.531***	0.547***	1.810**	0.566***	0.583***	1.859**	0.526***	0.540***	1.766**
	(13.72)	(13.22)	(2.35)	(14.62)	(14.23)	(2.49)	(13.98)	(13.39)	(2.31)
CEN	0.186***	0.177***	0.106**	0.282***	0.265***	0.179***	0.183***	0.169***	0.0910*
	(4.34)	(3.91)	(2.07)	(5.78)	(5.39)	(3.28)	(4.24)	(3.69)	(1.71)
Ctrls		X	X		X	X		X	X
SUE·Ctrls			X			X			X
Obs.	252,184	226,106	226,106	252,184	226,106	226,106	252,184	226,106	226,106
Adj. R^2 (%)	0.2	0.5	0.7	0.2	0.5	0.7	0.2	0.5	0.7

This table reports the regression results of cumulative abnormal returns (CAR) on the centrality of the firm's headquarters location. Panels A, B, and C correspond to the CARs for the announcement period (CAR[0, 1]) and the post-announcement periods (CAR[2, 40] and CAR[2, 61*]), respectively. CEN is the decile rank of the centrality of a firm's headquarters county, measured by degree centrality, eigenvector centrality, or information centrality. SUE is the decile rank of standardized unexpected earnings. All county- and firm-level control variables (lagged) and industry and time fixed effects listed in Section 1.2 and their interactions with SUE are included. Standard errors are two-way clustered by firm and announcement date, and the resultant t-statistics are shown in parentheses. *p <.1; **p <.05; ***p <.01.

Figure 3: Table 2, Panel C: Long-window post-announcement drift

5.3 Tables 3 and 4: Volatility and volume dynamics

Read:

- Table 3 shows negative centrality coefficients for volatility persistence.
- Table 4 shows positive centrality coefficients for immediate abnormal volume, post-announcement abnormal volume, and volume persistence.
- The contrast is important: centrality reduces return and volatility persistence but increases volume persistence.

Economic message. Information gets into prices faster, but social discussion keeps investors trading. This is the first core evidence for social churning.

Table 3
Centrality and volatility persistence

	Degree Centrality		Eigenvector Centrality		Information Centrality	
	(1)	(2)	(3)	(4)	(5)	(6)
CEN	−0.178***	−0.059***	−0.193***	−0.072***	−0.174***	−0.061***
	(−9.15)	(−3.58)	(−9.96)	(−4.31)	(−8.89)	(−3.57)
|SUE|	−0.101***	0.015	−0.103***	0.014	−0.102***	0.014
	(−8.92)	(1.30)	(−9.09)	(1.25)	(−8.96)	(1.29)
Ctrls		X		X		X
Obs.	249,426	223,698	249,426	223,698	249,426	223,698
Adj. R^2	0.2%	6.8%	0.2%	6.8%	0.2%	6.8%

This table reports the regression of volatility persistence on the centrality of the firm's headquarters location. The dependent variable, $d_{|R|}$, is the persistence parameter of the absolute returns series over the [0, 61*] window. CEN is the decile rank of the centrality of a firm's headquarters county, measured by degree centrality, eigenvector centrality, or information centrality. |SUE| is the decile rank of absolute standardized unexpected earnings. All county- and firm-level control variables (lagged) and industry and time fixed effects listed in Section 1.2 are included. Coefficients are multiplied by 100. Standard errors are two-way clustered by firm and announcement date, and the resultant t-statistics are shown in parentheses. *p <.1; **p <.05; ***p <.01.

(a) Table 3: Centrality and volatility persistence

Table 4
Centrality and trading volume

	LNVOL[0, 1]			LNVOL[2, 61*]			d_{VOL}		
	DC (1)	EC (2)	IC (3)	DC (4)	EC (5)	IC (6)	DC (7)	EC (8)	IC (9)
CEN	0.846***	1.018***	1.014***	0.062*	0.130***	0.082**	0.308***	0.369***	0.344***
	(5.56)	(6.60)	(6.41)	(1.74)	(3.37)	(2.17)	(10.75)	(12.69)	(11.50)
|SUE|	1.602***	1.614***	1.608***	0.833***	0.836***	0.834***	0.027*	0.031**	0.028**
	(19.03)	(19.21)	(19.09)	(18.33)	(18.38)	(18.34)	(1.86)	(2.15)	(1.96)
Obs.	233,218	233,218	233,218	232,687	232,687	232,687	205,779	205,779	205,779
Adj. R^2 (%)	4.4	4.4	4.4	2.8	2.8	2.8	17.6	17.7	17.6

This table reports the regression of trading volume on the centrality of the firm's headquarters location. In columns 1–3 and 4–6 the dependent variables are LNVOL[0, 1] and LNVOL[2, 61*], the average daily abnormal trading volume during the announcement window and the post-announcement window, respectively. In columns 7–9, the dependent variable is d_{VOL}, the persistent parameter of the daily abnormal volume over the [0, 61*] window. CEN is the decile rank of the centrality of a firm's headquarters county, measured by degree centrality (DC), eigenvector centrality (EC), or information centrality (IC). |SUE| is the decile rank of absolute standardized unexpected earnings. All county- and firm-level control variables (lagged) and industry and time fixed effects listed in Section 1.2 are included. Coefficients are multiplied by 100. Standard errors are two-way clustered by firm and announcement date, and the resultant t-statistics are shown in parentheses. *p <.1; **p <.05; ***p <.01.

(b) Table 4: Centrality and trading volume

5.4 Tables 5 and 6: StockTwits discussion and disagreement

Read:

- Table 5 shows that centrality increases new messages and replies during the announcement window.
- Post-announcement new messages fall for high-centrality firms, consistent with faster initial diffusion.
- Post-announcement replies rise strongly for high-centrality firms, consistent with continuing discussion.

- Table 6 shows that centrality is positively related to disagreement and disagreement persistence.

Economic message. Social networks do not merely spread facts. They also create continuing conversations and disagreement about how to interpret the facts.

Table 5
Centrality and StockTwits mentions

A. New messages

	ANM[0, 1]			ANM[2, 61*]		
	DC (1)	EC (2)	IC (3)	DC (4)	EC (5)	IC (6)
CEN	0.34**	0.42**	0.40**	−0.07***	−0.09***	−0.07**
	(2.07)	(2.56)	(2.37)	(−2.82)	(−3.00)	(−2.51)
|SUE|	2.69***	2.70***	2.70***	0.44**	0.43**	0.44**
	(5.37)	(5.40)	(5.39)	(2.40)	(2.39)	(2.40)
Obs.	35,940	35,940	35,940	35,940	35,940	35,940
Adj. R^2 (%)	36.8	36.8	36.8	9.7	9.7	9.7

B. Reply messages

	ARM[0, 1]			ARM[2, 61*]		
	DC (1)	EC (2)	IC (3)	DC (4)	EC (5)	IC (6)
CEN	0.83***	1.16***	0.86***	1.08***	1.51***	1.18***
	(3.42)	(4.68)	(3.39)	(4.03)	(5.51)	(4.22)
|SUE|	1.97**	2.00**	1.97**	3.01***	3.06***	3.02***
	(2.27)	(2.31)	(2.28)	(3.35)	(3.40)	(3.36)
Obs.	34,326	34,326	34,326	34,326	34,326	34,326
Adj. R^2 (%)	27.1	27.1	27.1	28.8	28.9	28.8

This table reports the regression of abnormal StockTwits message activities on the centrality of the firm's headquarters location. Panels A and B present the results for abnormal new messages and abnormal replies, respectively. Abnormal New Messages, ANM[0, 1] and ANM[2, 61*], are the abnormal average daily number of new messages for the [0, 1] and [2, 61*] windows, respectively, relative to its preannouncement average. Similarly, ARM[0, 1] and ARM[2, 61*] are the abnormal average daily reply messages for the corresponding windows. CEN is the decile rank of the centrality of a firm's headquarters county, measured by degree centrality (DC), eigenvector centrality (EC), or information centrality (IC). |SUE| is the decile rank of absolute standardized unexpected earnings. All county- and firm-level control variables (lagged) and industry and time fixed effects listed in Section 1.2 are included. Standard errors are two-way clustered by firm and announcement date and the

(a) Table 5: StockTwits messages and replies

Table 6
Centrality and StockTwits disagreement

A. Message disagreement

	DIS[0, 1]			DIS[2, 61*]			d_{DIS}		
	DC (1)	EC (2)	IC (3)	DC (4)	EC (5)	IC (6)	DC (7)	EC (8)	IC (9)
CEN	0.521	1.150***	0.516	1.448***	2.196***	1.528***	0.388***	0.490***	0.423***
	(1.28)	(2.61)	(1.19)	(3.66)	(5.40)	(3.71)	(3.67)	(4.51)	(3.87)
|SUE|	−0.101	−0.090	−0.101	−0.021	−0.008	−0.019	0.058	0.060	0.059
	(−0.41)	(−0.37)	(−0.42)	(−0.09)	(−0.03)	(−0.08)	(0.82)	(0.85)	(0.82)
Obs.	21,460	21,460	21,460	30,105	30,105	30,105	26,562	26,562	26,562
Adj. R^2 (%)	10.4	10.4	10.4	18.8	18.9	18.8	8.3	8.4	8.3

B. Reply disagreement

	DIS[2, 61*]			d_{DIS}		
	DC (1)	EC (2)	IC (3)	DC (4)	EC (5)	IC (6)
CEN	1.336***	2.080***	1.520***	0.374***	0.479***	0.384***
	(3.41)	(5.21)	(3.79)	(3.56)	(4.45)	(3.54)
|SUE|	0.138	0.153	0.141	0.004	0.006	0.004
	(0.59)	(0.65)	(0.60)	(0.05)	(0.07)	(0.05)
Obs.	28,895	28,895	28,895	25,591	25,591	25,591
Adj. R^2 (%)	19.6	19.7	19.7	7.4	7.5	7.4

This table reports the regression of disagreement of StockTwits messages on the centrality of the firm's headquarters location. Panel A corresponds to disagreement across all messages, and panel B corresponds to disagreement across relies. DIS[0, 1] and DIS[2, 61*] refer to the average abnormal daily disagreement over the [0, 1] and [2, 61*] windows, respectively, compare to the preannouncement mean. d_{DIS} is the persistence parameter of disagreement, measured over the [0, 61*] window. CEN is the decile rank of the centrality of a firm's headquarters county based on the degree centrality (DC), eigenvector centrality (EC), or information centrality (IC). |SUE| is the decile rank of absolute standardized unexpected earnings. All county- and firm-level control variables (lagged) and industry and time fixed effects listed in Section 1.2 are included. Standard errors are two-way clustered by firm and announcement date, and the resultant t-statistics are shown in parentheses. $^*p$ <.1; $^{**}p$ <.05; $^{***}p$ <.01.

(b) Table 6: StockTwits disagreement

5.5 Tables 7 and 8: Influencers and Google Search

Read:

- Table 7 shows that initial posts by high-centrality StockTwits users predict more later replies and more later disagreement.
- Influencer posts are associated with lower volatility persistence but higher volume persistence.
- Table 8 shows that high-centrality firms receive more Google Search attention and more persistent search interest.

Economic message. The social churning mechanism appears both within StockTwits and in broader investor attention measured by Google Search.

Table 7
Influencer posts, replies, and the persistence of volatility and volume

	(1) ARM[2, 61*]	(2) DIS[2, 61*]	(3) $d_{\|R\|}$	(4) d_{VOL}
INFL[0, 1]	0.019***	0.015***	−0.114*	0.662***
	(4.19)	(3.16)	(−1.86)	(9.45)
ANM[0, 1]	0.398***	−0.074****	0.498**	0.900***
	(12.59)	(−5.18)	(2.17)	(3.48)
|SUE|	0.005***	−0.002	0.035	0.006
	(2.89)	(−0.60)	(1.30)	(0.23)
Obs.	34,232	20,917	35,940	35,940
Adj. R^2 (%)	46.7	42.8	7.4	13.2

This table reports the results of the regression analysis of StockTwits influencer posts and the subsequent messaging activities as well as the volatility and volume persistence. The dependent variables for columns 1 and 2 are ARM[2, 61*] and DIS[2, 61*], the abnormal number of replies and the abnormal daily message disagreement for the post-announcement window of [2, 61*], respectively. For columns 3 and 4, the dependent variables are volatility persistence ($d_{|R|}$) and volume persistence (d_{VOL}), respectively. The independent variables are INFL[0, 1] and ANM[0, 1], the average sender centrality of new messages and abnormal new messages for the [0, 1] window, respectively, and |SUE|, the decile rank of absolute standardized unexpected earnings. All county- and firm-level control variables (lagged) and industry and time fixed effects listed in Section 1.2 are included. Standard errors are two-way clustered by firm and announcement date, and the resultant t-statistics are shown in parentheses. $^*p$ <.1; $^{**}p$ <.05; $^{***}p$ <.01.

(a) Table 7: Influencer posts and later dynamics

Table 8
Centrality and Google searches

	ASV[0, 1]			ASV[2, 61*]			d_{ASV}		
	DC (1)	EC (2)	IC (3)	DC (4)	EC (5)	IC (6)	DC (7)	EC (8)	IC (9)
CEN	0.280**	0.659***	0.366***	0.037	0.056**	0.039	0.368***	0.297**	0.356***
	(2.11)	(4.64)	(2.65)	(1.43)	(2.04)	(1.43)	(3.00)	(2.43)	(2.82)
|SUE|	0.130**	0.139**	0.132**	0.087***	0.087***	0.087***	−0.045	−0.044	−0.044
	(2.01)	(2.16)	(2.05)	(3.72)	(3.75)	(3.73)	(−1.28)	(−1.26)	(−1.26)
Obs.	115,452	115,452	115,452	113,512	113,512	113,512	111,871	111,871	111,871
Adj. R^2 (%)	1.8	1.9	1.8	1.7	1.7	1.7	11.9	11.9	11.9

This table reports the regression of investor attention on the centrality of the firm's headquarters location. The dependent variable for columns 1–3 is ASV[0, 1], the abnormal Google searches for the announcing stock in the announcement window. The dependent variable for columns 4–6 is ASV[2, 61*], the abnormal Google searches in the post-announcement window. For columns 7–9, the dependent variable is d_{ASV}, the persistence of Google searches over the [0, 61*] window. CEN is the decile rank of the centrality of a firm's headquarters county based on the degree centrality (DC), eigenvector centrality (EC), or information centrality (IC), respectively. |SUE| is the decile rank of absolute standardized unexpected earnings. All county- and firm-level control variables (lagged) and industry and time fixed effects listed in Section 1.2 are included. Coefficients are multiplied by 100. Standard errors are two-way clustered by firm and announcement date, and the resultant t-statistics are shown in parentheses. $^*p$ <.1; $^{**}p$ <.05; $^{***}p$ <.01.

(b) Table 8: Centrality and Google searches

5.6 Tables 9 and 10: Household trading and Facebook versus StockTwits centrality

Read:

- Table 9 shows that households more socially connected to the announcing firm's county trade more frequently, more often, and in larger size.
- The post-announcement net-profit coefficient is negative, while the trading-cost coefficient is positive.
- Table 10 shows that StockTwits centrality is not significantly associated with return reactions, whereas Facebook centrality remains informative.

Economic message. Broad offline social networks appear more important for price diffusion, while online platform activity is more transient. At the household level, social diffusion can increase excessive trading.

Table 9
Social ties and household trading

A. Trading activities

	Trading Indicator		Number of Trades		Relative Trade Size	
	[0, 1] (1)	[2, 61*] (2)	[0, 1] (3)	[2, 61*] (4)	[0, 1] (5)	[2, 61*] (6)
RSCI	0.015***	0.162***	0.018***	0.321***	0.005***	0.143***
	(3.08)	(9.61)	(3.43)	(8.45)	(4.56)	(8.88)
|SUE|	0.056***	0.379***	0.063***	0.740***	0.011***	0.184***
	(4.19)	(6.13)	(4.18)	(5.17)	(4.55)	(5.42)
Obs.	3,916,866	3,916,866	3,916,866	3,916,866	3,916,866	3,916,866
Adj. R^2 (%)	1.1	6.3	1.2	6.6	1.5	6.0

B. Trading profits

	[0, 1]			[2, 61*]		
	Profit^{net} (1)	Profit^{gross} (2)	Cost (3)	Profit^{net} (4)	Profit^{gross} (5)	Cost (6)
RSCI	−0.007	−0.002	0.005***	−0.151**	0.009	0.178***
	(−1.48)	(−0.45)	(2.79)	(−2.31)	(0.15)	(6.76)
|SUE|	−0.032**	−0.017	0.014***	−0.687***	−0.404***	0.254***
	(−2.42)	(−1.56)	(3.67)	(−3.71)	(−2.67)	(5.17)
Obs.	3,916,866	3,916,866	3,916,866	3,916,866	3,916,866	3,916,866
Adj. R^2 (%)	0.2	0.1	1.0	1.4	1.0	3.8

This table analyzes households' trading activities and profits following earnings announcements. In panel A, the dependent variable is the trading activity of a household on the announcing stock for a given window, measured three ways: (1) a trading indicator, (2) the number of trades, or (3) relative trade size. For panel B, the dependent variable is the profit of a household from trading the announcing stock for a given window, with a negative value corresponding to a loss. Profit^{net} and Profit^{gross} are the net and gross profit for a household, respectively. Cost is the trading costs. All Profit and Cost measures are scaled by the household's beginning-of-month stock portfolio value before the announcement and multiplied by 10^4. RSCI (in logarithm) is relative social connectedness between the locations of the firm and the household. |SUE| is the decile rank of absolute standardized unexpected earnings. We include time indicator variables, lagged firm and household control

(a) Table 9: Social ties and household trading

Table 10
Facebook versus StockTwits centrality

A. Return reactions

	CAR[0, 1]				CAR[2, 61*]			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
SCEN·SUE	0.00870	0.00833	0.00784	0.00822	0.00872	0.00941	0.00929	0.00940
	(1.36)	(1.31)	(1.23)	(1.29)	(0.53)	(0.57)	(0.56)	(0.57)
DC·SUE		0.0111*				−0.0203		
		(1.72)				(−1.28)		
EC·SUE			0.0126*				−0.0134	
			(1.85)				(−0.82)	
IC·SUE				0.0126*				−0.0208
				(1.86)				(−1.28)
SCEN	−0.157***	−0.154***	−0.151***	−0.153***	−0.147	−0.153	−0.158	−0.154
	(−3.85)	(−3.77)	(−3.69)	(−3.75)	(−1.24)	(−1.29)	(−1.34)	(−1.30)
SUE	2.648***	2.461***	2.605***	2.506***	1.928	2.269	1.976	2.161
	(4.36)	(4.00)	(4.28)	(4.10)	(1.13)	(1.38)	(1.17)	(1.29)
Ctrls(· SUE)	X	X	X	X	X	X	X	X
Obs.	47,335	47,335	47,335	47,335	47,191	47,191	47,191	47,191
Adj. R^2 (%)	3.9	3.9	3.9	3.9	1.6	1.6	1.6	1.6

B. Volume reactions

	LNVOL[0, 1]				LNVOL[2, 61*]			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
SCEN	1.470***	1.430***	1.405***	1.423***	−1.091***	−1.099***	−1.115***	−1.101***
	(6.78)	(6.60)	(6.48)	(6.57)	(−8.61)	(−8.65)	(−8.76)	(−8.67)
DC		0.694**				0.145*		
		(2.56)				(1.85)		
EC			0.772***				0.298***	
			(2.82)				(3.84)	
IC				0.791***				0.182**
				(2.76)				(2.18)
|SUE|	1.114***	1.128***	1.131***	1.131***	0.719***	0.722***	0.725***	0.723***
	(7.78)	(7.90)	(7.93)	(7.93)	(8.94)	(8.97)	(9.02)	(8.99)
Ctrls	X	X	X	X	X	X	X	X
Obs.	48,714	48,714	48,714	48,714	48,651	48,651	48,651	48,651
Adj. R^2 (%)	5.1	5.1	5.1	5.1	3.0	3.0	3.0	3.0

This table reports the regression results of return and volume reactions on StockTwits centrality and Facebook centrality, presented in panels A and B, respectively. SCEN is the decile rank of the StockTwits centrality of a firm, measured by the total number of messages mentioning the firm's stock ticker on the social media platform in the past 3 months. CEN is the decile rank of the centrality of a firm's headquarters county, measured by degree centrality (DC), eigenvector centrality (EC), or information centrality (IC). |SUE| is the decile rank of absolute earnings surprises. All county- and firm-level control variables and industry and time fixed effects listed in Section 1.2 are included. For panel A, the control variables are also interacted with SUE. Standard errors

(b) Table 10: Facebook versus StockTwits centrality

5.7 Figure A1: Post-announcement dynamics

Read:

- Panel A shows that return drift declines over post-announcement windows for high-centrality firms.
- Panel B shows volume effects persist across post-announcement windows.
- Panels C and D show the contrast between new messages and replies.

Economic message. The dynamic pattern summarizes the paper: faster price adjustment, persistent volume, and persistent social discussion.

A. Figure and Variable List

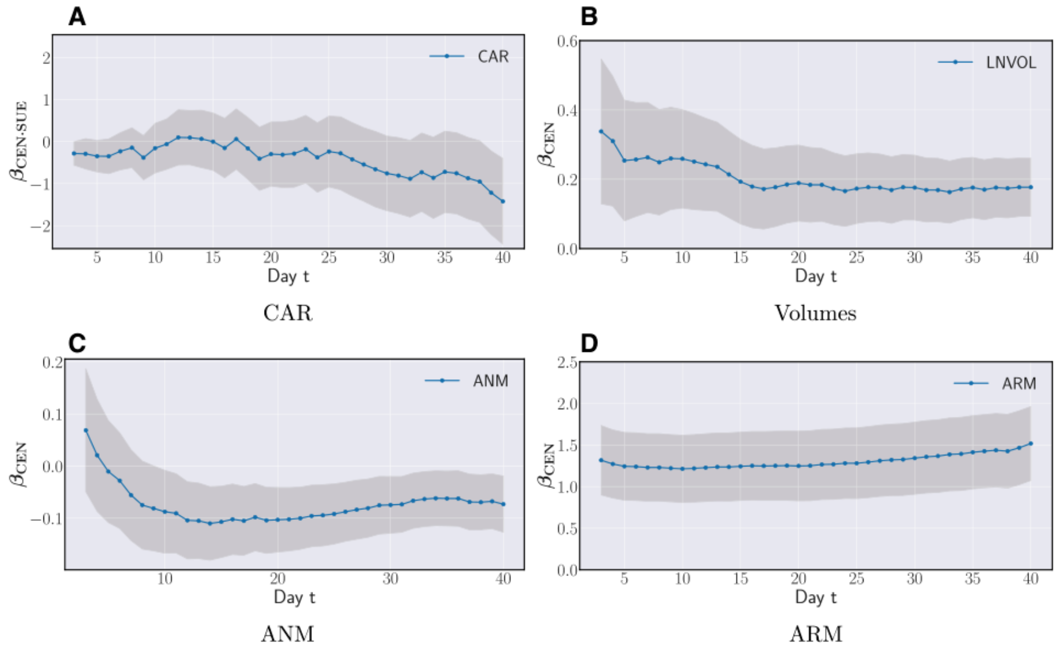

Figure A1
Centrality and post-announcement dynamics
This figure summarizes the regression coefficients of eigenvector centrality in explaining returns, trading volume and message activities over the post-announcement windows [2, 3], [2, 4], ..., [2, 40]. Panels A through D

Figure 8: Figure A1: Centrality and post-announcement dynamics

6 Short summary

- The paper examines how social network centrality affects market reactions to earnings announcements.
- Firm centrality is measured by the Facebook connectedness centrality of the firm's headquarters county.
- High-centrality firms have stronger immediate price reactions to earnings surprises.
- High-centrality firms have weaker post-earnings-announcement drift.
- High-centrality firms have faster volatility decay.
- However, high-centrality firms have higher and more persistent trading volume.
- The paper explains this contrast with the social churning hypothesis.
- StockTwits evidence confirms that high-centrality news generates more replies and disagreement.
- Google Search evidence shows that attention remains elevated after high-centrality announcements.
- Household data show that stronger social ties predict more trading and lower net performance.

7 Conclusion

7.1 Core conclusion

Hirshleifer, Peng, and Wang show that social networks affect how public earnings news is incorporated into asset prices. When a firm's local investor base is more central in the social network, the market reacts to earnings news faster and post-announcement drift is weaker. This supports the view that investor attention spreads through social connections.

7.2 Economic intuition

The main economic intuition is a tension between information diffusion and belief churning. Fast diffusion reduces the set of inattentive investors and improves price discovery. But the same social interactions that spread news also spread interpretations, opinions, and disagreement. That continuing discussion generates persistent trading volume even after prices and volatility have largely adjusted.

7.3 Takeaway

The paper moves the social-finance literature beyond the question of whether social interactions matter. It shows that social networks can be good for markets and bad for individual investors at the same time: they speed up price discovery but also encourage attention, disagreement, excessive trading, trading costs, and losses.